

Assignment 3

1. Determine V_0 for the circuit given in Fig. 1. Assume zero forward bias voltage for green LED to turn on. Will the green LED turn on?? Justify your explanation in few words.

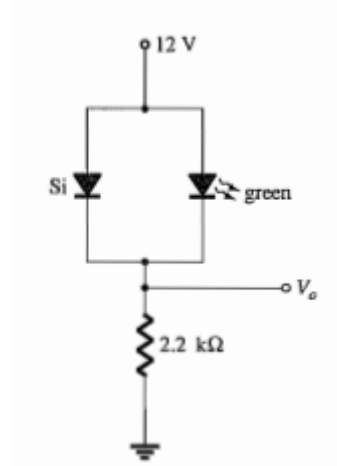


Fig. 1

2. Determine the output Voltage V_0 for the given circuit in Fig. 2. Can you comment which digital circuit (logic gate) is getting replicated through your calculation and analysis.

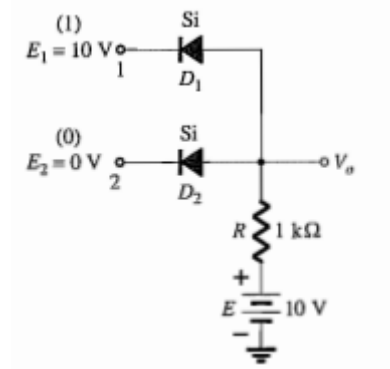


Fig. 2

3. Sketch the output waveform for the given circuit in Figures 3 (a) and (b) and mention the amplitude, time instant properly for one full cycle.

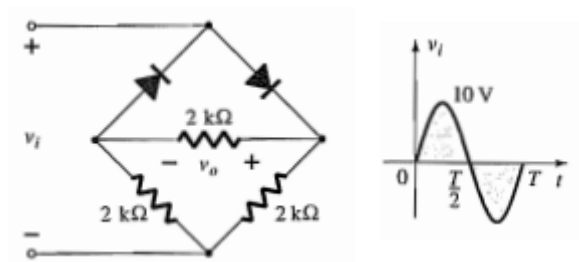


Fig. 3 (a) & (b)

4. Determine the range of input voltage V_i for which Zener diode will maintain ON state. V_Z and I_{ZM} represent reverse breakdown voltage and maximum current through the Zener diodes respectively. Refer Fig. 4 for this question.

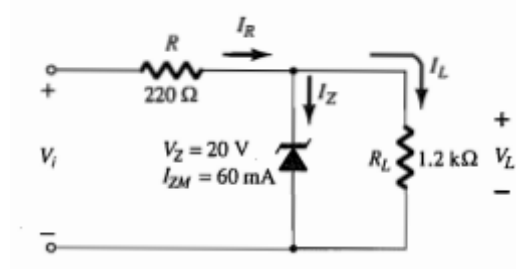


Fig. 4

5. Determine the reference voltages provided by the network of Fig. 5, which uses a white LED to indicate that the power is on. What is the level of current through the LED and the power delivered by the supply? How does the power absorbed by the LED compare to that of the 6-V Zener diode?

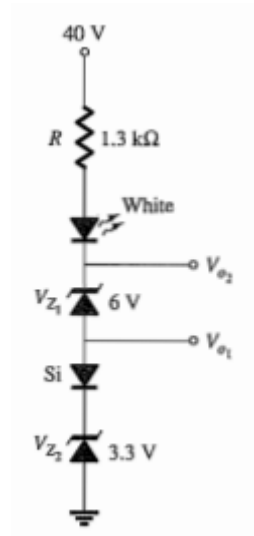


Fig. 5

6. Draw the output waveform for the circuit given in Fig. 6(a) for the shown input in Fig. 6 (b) Mention amplitude and time instant properly.

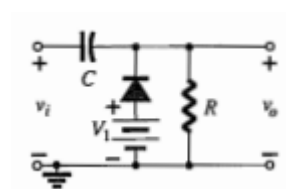


Fig. 6 (a)

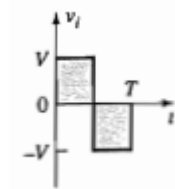


Fig. 6 (b)

7. Determine V_0 for the network of Fig. 7(a) for the input indicated in Fig. 7(b).

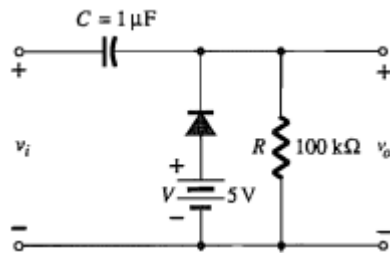


Fig. 7(a)

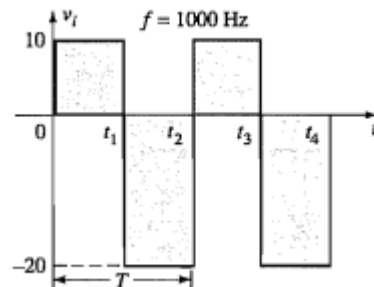


Fig. 7(b)

8. Calculate the output voltage for the circuit of Fig. 8. The inputs are $V_1 = 50 \sin(1000t)$ mV and $V_2 = 10 \sin(3000t)$ mV.

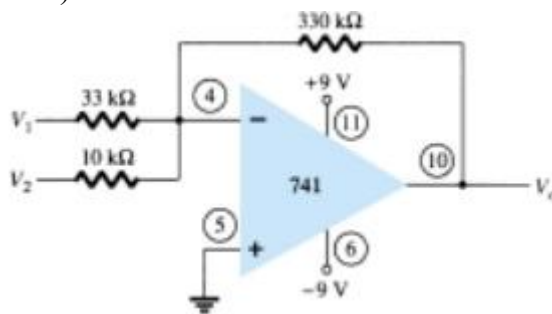


Fig. 8

9. Calculate the output voltage of a three-stage op-amp amplifier with gains of +10, -18, and -27. The value of feedback resistor is $270 \text{ k}\Omega$ for all three circuits. What output voltage will result for an input of $150 \mu\text{V}$? (Refer Fig. 9).

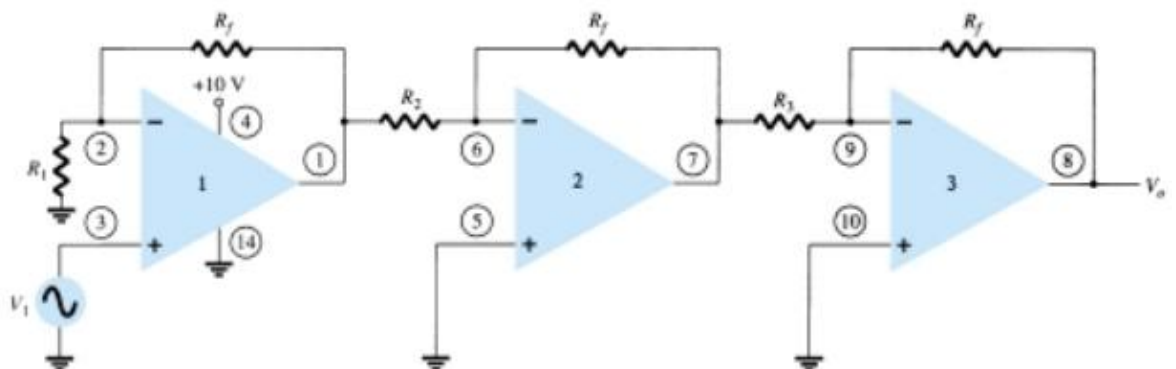


Fig. 9

10. Find v_o and i_o in the op-amp circuit shown in Fig. 10.

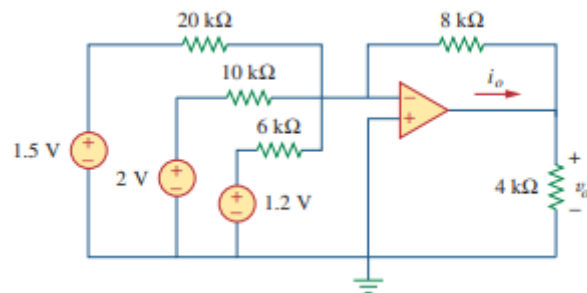


Fig. 10

11. Calculate i_o for the given circuit in Fig. 11.

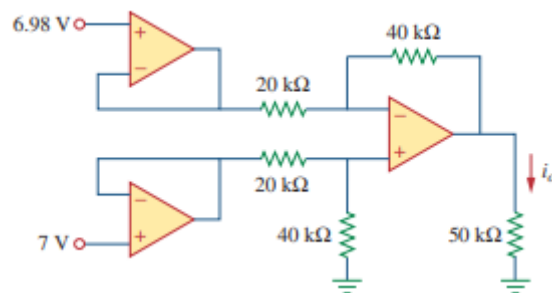


Fig. 11

12. Calculate v_o in the circuit of Fig. 12.

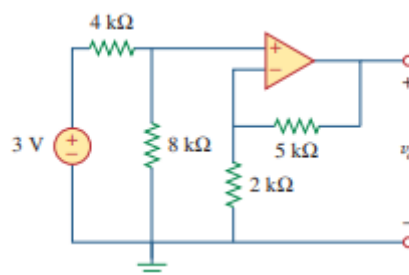


Fig. 12

13. Determine the output voltage for the circuit of Fig. 13 with components $R_f = 1\text{ M}\Omega$, $R_1 = 100\text{ k}\Omega$, $R_2 = 50\text{ k}\Omega$, and $R_3 = 500\text{ k}\Omega$.

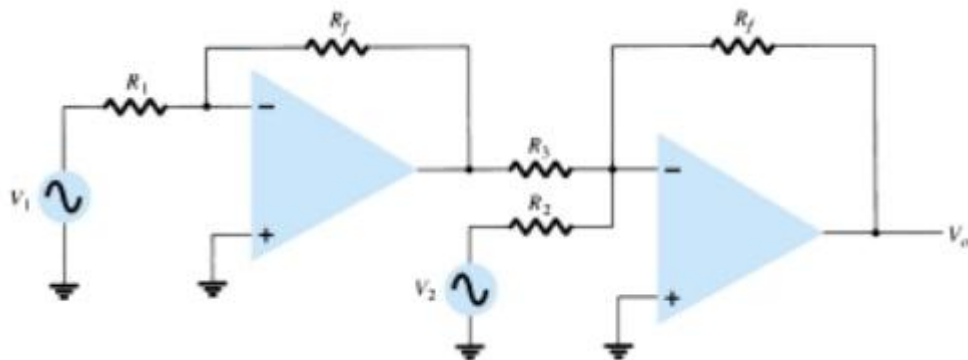


Fig. 13

14. Calculate the output voltage expression for the circuit of Fig. 14.

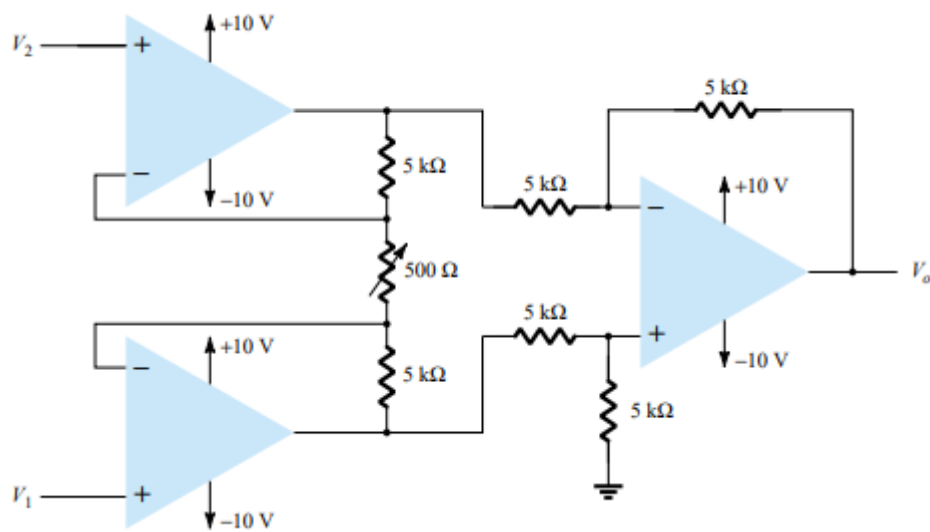


Fig. 14